A COMPLETELY DECOMPOSED FINITELY PRESENTED MAP ADMITS A FINITE COVER OF THE TARGET BY LOCALLY CLOSED PIECES OF FINITE PRESENTATION OVER WHICH A SECTION EXISTS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 23186-23204 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 23206-23237, the complete author proof (30 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The reference block cites \cite[Lemma 2.1.2]{EHIK}; a bibliography.tex with the snapshot my.bib EHIK entry is attached (precedent L16058/L16018).

Editorial note. The statement's reference environment cites \cite[Lemma 2.1.2]{EHIK}; a bibliography.tex holding the snapshot my.bib EHIK entry is attached (precedent L16058/L16018).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Directed-limit presentation of residue fields by finitely presented subalgebras, the Limits chapter lifting proposition for morphisms out of finite presentation targets, Chevalley's constructibility theorem, and the constructible-topology compactness of qcqs schemes.

Source reference. Original lemma 01ZC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23237–23237. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 054J: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23237–23237. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 094L: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23237–23237. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03GI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23237–23237. Cited by the retained statement or

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0901: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23237–23237. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. [1, Lemma 2.1.2] *Let $f : X \to Y$ be a morphism of schemes. Assume f is completely decomposed, f is locally of finite presentation, and Y is quasi-compact and quasi-separated. Then there exist $n \geq 0$ and morphisms $Z_i \to Y$, $i = 1, \ldots, n$ with the following properties*

(1) *$\coprod Z_i \to Y$ is surjective,*
(2) *$Z_i \to Y$ is an immersion for all i,*
(3) *$Z_i \to Y$ is of finite presentation for all i, and*
(4) *the base change $X \times_Y Z_i \to Z_i$ has a section for all i.*

Proof. Let $y \in Y$. By assumption there is a morphism $\sigma : \mathrm{Spec}(\kappa(y)) \to X$ over Y. We can write $\mathrm{Spec}(\kappa(y))$ as a directed limit of affine schemes Z over Y such that $Z \to Y$ is an immersion of finite presentation. Namely, choose an affine open $y \in \mathrm{Spec}(A) \subset Y$ and say y corresponds to the prime ideal $\mathfrak{p}$ of A. Then $\kappa(\mathfrak{p})$ is the filtered colimit of the rings $(A/I)_f$ where $I \subset \mathfrak{p}$ is a finitely generated ideal and $f \in A$, $f \notin \mathfrak{p}$. The morphisms $Z = \mathrm{Spec}((A/I)_f) \to Y$ are immersions of finite presentation; quasi-compactness of $Z \to Y$ follows as Y is quasi-separated, see Schemes, Lemma 03GI. By Limits, Proposition 01ZC for some such Z there is a morphism $\sigma' : Z \to X$ over Y agreeing with σ on the spectrum of $\kappa(\mathfrak{p})$. Since σ' is a morphism over Y, we obtain a section of the projection $X \times_Y Z \to Z$

We conclude that Y is the union of the images of immersions $Z \to Y$ of finite presentation such that $X \times_Y Z \to Z$ has a section. Since the image of $Z \to Y$ is constructible (Morphisms, Lemma 054J) and since Y is compact in the constructible topology (Properties, Lemma 094L and Topology, Lemma 0901), we see that a finite number of these suffice. □

References

[1] Elmanto, Elden and Hoyois, Marc and Iwasa, Ryomei and Kelly, Shane. *Cdh descent, cdarc descent, and Milnor excision*. Math. Ann. 379 (2021), no. 3-4, 1011–1045.